IP PROJECT

BANK MANAGEMENT SYSTEM

FOR AISSCE 2025 EXAMINATION

AS PART OF INFORMATICS PRACTICES COURSE(65)

DEPARTMENT OF INFORMATICS PRACTICES

SRI CHAITANYA TECHNO SCHOOL

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ABSTRACT

This is a small project on the Bank Management System. This system allows the customer to do various operations . We have designed an interface to store all details of customers in a database (csv files) using python.

The Bank System enables administrators to create an account by entering credentials (For example- username and password). By using this software, the details of the account, customer and amount can be stored. In addition, we can insert new customer details, select existing customer details, view graphical interpretation. This program is made from the point of view of a bank manager

SYSTEM REQUIREMENTS

Hardware Components:

1. VGA Monitor
2. Qwerty keyboard
3. 2GB RAM
4. 2.6 GHz Processor
5. Graphics card

Software Components:

1. Windows 10
2. Python 3.13 with suitable modules

CODING

lp_projectfinal.py - C:\Users\Vishveshwar\Desktop\Vaishnavi\lp_projectfinal.py (3.13.1)

File Edit Format Run Options Window Help

```
The management team is responsible for setting long-term goals, making decisions on financial resources, mitigating risks, complying with regulations, enhancing cus
Effective bank management is vital for the stability and growth of a financial institution.\n The main objective is to build an organic and optimal interaction syst
            print("+" + "_" * 120 + "+")

        elif choice == 1:
            print("Account Operations")
            print("+-------------------------------+")
            print("|OPTION|      DESCRIPTION       |")
            print("+-------------------------------+")
            print("|  1   | Open a new account     |")
            print("|  2   | Modify existing account|")
            print("|  3   | Close an account       |")
            print("|  4   | Check account balance  |")
            print("|  5   | Exit                   |")
            print("+-------------------------------+")

            account_option = int(input("Choose your desired option: "))

            if account_option == 1:
                print("\tOpening a new account")
                new_customer = input("Is this a new customer? If yes go back to the main menu and register otherwise, press any y and continue : ")
                if new_customer in 'Yy':
                    customer_name = input("Enter customer name: ")
                    # Assuming Account and Customer are pandas DataFrames
                    customer_id = Customer[Customer.loc[:, "customer_name"] == customer_name]["Customer_name"]
                    print(customer_id)
                    if customer_id.empty:
                        print("This customer is not registered. Go to the main menu and register first.")
                    else:
                        account_id = Account.index
                        account_id = account_id[-1] + 1
                        print(f"Congratulations! The auto-generated Account number is {account_id}")
                        account_type = input("Enter account type (saving/current):\n For Saving 'S' and for Current 'C': ")
                        if account_type in "Ss":
                            account_type = "Savings"
                            balance = float(input("Enter initial balance: Rs."))
                            Account.loc[account_id] = [customer_name, account_type, balance]
                            Account.to_csv("Account.csv")
                            print("New account opened successfully")
                            print(Account.loc[account_id])
                        else:
```

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```
                        print(Account.loc[account_id])
                    else:
                        print("Wrong entry.")
                else:
                    print("For registration as a new customer, go back to the main menu.")

            elif account_option == 2:
                print("\tModifying an existing account")
                account_id = int(input("Enter account ID: "))
                if account_id not in Account.index:
                    print("No such account exists. Check the account ID.")
                    print("Press enter to continue.")
                else:
                    print(Account.loc[account_id])
                    balance = float(input("Enter new balance in Rs: "))
                    Account.loc[account_id, "Balance"] = balance
                    Account.to_csv("Account.csv")
                    print("Account data changed successfully.")
                    input("Press enter to continue.")

            elif account_option == 3:
                print("\tClosing an existing account")
                account_id = int(input("Enter account ID: "))
                if account_id not in Account.index:
                    print("No such account exists. Check the account ID.")
                    print("Press enter to continue.")
                else:
                    print(Account.loc[account_id])
                    close_confirm = input("Do you want to close this account? (Y/N): ")
                    if close_confirm.lower() in 'Yy':
                        Account.drop(account_id, axis=0, inplace=True)
                        Account.to_csv("Account.csv")
                        print("Account closed successfully.")
                        input("Press enter to continue.")
                    else:
                        print("Account not closed.")
                        input("Press enter to continue.")

            elif account_option == 4:
                print("Checking Account balance")
                account_id = int(input("Enter account ID: "))
```

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```
                print("Checking Account balance")
                account_id = int(input("Enter account ID: "))
                if account_id not in Account.index:
                    print("No such account exists. Check the account ID.")
                    print("Press enter to continue.")
                else:
                    print(f"Account balance: Rs {Account.loc[account_id, 'Balance']}")
                    input("Press enter to continue.")

            elif account_option == 5:
                print("Exiting successfully")
                break

        elif choice == 2:
            print("Customer Operations")
            print("+--------------------------------+")
            print("|OPTION|      DESCRIPTION       |")
            print("+--------------------------------+")
            print("|  1   | Add a new customer      |")
            print("|  2   | Modify an existing customer |")
            print("|  3   | Remove a customer       |")
            print("|  4   | View customer information   |")
            print("|  5   | Graphical representation |")
            print("|  6   | Exit                    |")
            print("+--------------------------------+")

            customer_option = int(input("Choose your desired option: "))

            if customer_option == 1:
                print("\tAdding a New Customer")
                customer_name = input("Enter customer name: ")
                customer_address = input("Enter customer address: ")
                customer_email = input("Enter customer email: ")
                customer_id = Customer.index
                customer_id = customer_id[-1] + 1
                customer_id = "C" + str(customer_id)
                print(f"Customer ID is {customer_id}")
                Customer.loc[customer_id] = [customer_name, customer_address, customer_email]
                Customer.to_csv("Customer.csv")
                print("New customer added successfully.")
                print(Customer.loc[customer_id])
```

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File Edit Format Run Options Window Help

```python
                print("New customer added successfully.")
                print(Customer.loc[customer_id])
                input("Press any key to continue.")
elif customer_option == 2:
    print("\tModifying an existing customer")
    customer_id = input("Enter customer ID: ")
    if customer_id not in Customer.index:
        print("No such customer exists. Check the customer ID.")
        input("Press any key to continue.")
    else:
        print(Customer.loc[customer_id])
        new_name = input("Enter new customer name: ")
        new_address = input("Enter new customer address: ")
        new_email = input("Enter new customer email: ")
        Customer.loc[customer_id] = [new_name, new_address, new_email]
        Customer.to_csv("Customer.csv")
        print("Customer data changed successfully.")
        input("Press any key to continue.")

elif customer_option == 3:
    print("\tRemoving existing customer")
    customer_id = input("Enter customer ID: ")
    if customer_id not in Customer.index:
        print("No such customer exists. Check the customer ID.")
        input("Press any key to continue.")
    else:
        print(Customer.loc[customer_id])
        remove_confirm = input("Do you want to remove this customer? (Y/N): ")
        if remove_confirm.lower() in 'Yy':
            Customer.drop(customer_id, axis=0, inplace=True)
            Customer.to_csv("Customer.csv")
            print("Customer removed successfully.")
            input("Press any number to continue.")
        else:
            print("Customer not removed.")
            input("Press any key to continue.")

elif customer_option == 4:
    print("Want to print a Specific customer press 1")
    print("Want to print Top 5 customer details press 2")
    print("Want to print bottom 5 customer details press 3")
```

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```python
    print("Want to print Top 5 customer details press 2")
    print("Want to print bottom 5 customer details press 3")
    print("Want to print only customer names and Phone numbers press 4")
    print("Want to print customer names in ascending order press 5")
    print("Want to print customer names in descending order press 6")
    CChoice = int(input("Enter your choice: "))

    if CChoice == 1:
        customer_id = input("Enter customer ID: ")
        if customer_id not in Customer.index:
            print("No such customer exists. Check the customer ID.")
            input("Press any key to continue.")
        else:
            print(Customer.loc[customer_id])
    elif CChoice == 2:
        print(Customer.head())
    elif CChoice == 3:
        print(Customer.tail())
    elif CChoice == 4:
        print(Customer[["Customer name", "Customer Address"]])
    elif CChoice == 5:
        print(Customer.sort_values(by="customer_Name"))
    elif CChoice == 6:
        print(Customer.sort_values(by="Customer_Name", ascending=False))
    input("Press enter to continue.")

elif customer_option == 5:
    print("Graphical representation of account data")
    print("+--------------------------------+")
    print("|OPTION|      DESCRIPTION       |")
    print("+--------------------------------+")
    print("|  1   | Bar Chart: Total balance by type")
    print("|  2   | Pie Chart: Balance distribution")
    print("|  3   | Line Chart: Balance over time")
    print("|  4   | Histogram: Balance distribution")
    print("+--------------------------------+")

    chart_type = int(input("Select the desired chart type: "))
    account_data = pd.read_csv("Account.csv")
    account_data.set_index(' Account_ID', inplace=True)
```

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```python
    account_data.set_index(' Account_ID', inplace=True)

    # Bar Chart
    if chart_type == 1:
        account_balance = account_data.groupby(" Account_Type")["Balance"].sum()
        account_balance.plot(kind="bar")
        plt.xlabel("Account Type")
        plt.ylabel("Total Balance")
        plt.title("Account Balances by Type")
        plt.xticks(rotation=45)
        plt.show()
        print("Bar chart generated: Shows total balance for each account type.")

    # Pie Chart
    elif chart_type == 2:
        account_balance = account_data.groupby("Account_Type")["Balance"].sum()
        account_balance.plot(kind="pie", autopct="%1.1f%%")
        plt.title("Account Balances by Type")
        plt.show()
        print("Pie chart generated: Shows total balance for each account type.")

    # Line Chart
    elif chart_type == 3:
        account_data[["Balance"]].plot(marker='o', linestyle='-')
        plt.xlabel("Account ID")
        plt.ylabel("Balance")
        plt.xticks(rotation=30)
        plt.legend(['Balance'])
        plt.show()
        print("Line chart generated: Represents the balance for each account ID over time")

    # Histogram
    elif chart_type == 4:
        account_data["Balance"].plot(kind="hist", bins=10)
        plt.xlabel("Balance")
        plt.ylabel("Frequency")
        plt.title("Distribution of account balances")
        plt.show()

elif choice == 3:
    print("        Transaction Operations    ")
```

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```python
elif choice == 3:
    print("        Transaction Operations    ")
    print("+----------------------------------------+")
    print("| Option |     Description               |")
    print("+----------------------------------------+")
    print("|   1    |   Deposit                     |")
    print("|   2    |   Withdraw                    |")
    print("|   3    |   Transfer                    |")
    print("|   4    |   View Transaction History    |")
    print("|   5    |   Exit                        |")
    print("+----------------------------------------+")

    transaction_option = int(input("Choose your Desired Option: "))

if transaction_option == 1:
    print("\tDepositing Money")
    account_id = int(input("Enter Account ID: "))
    if account_id not in Account.index:
        print("No Such Account Exists")
        print("Check the Account ID")
        input("Press Enter to continue: ")

    else:
        amount = float(input("Enter the deposit amount (in Rs): "))
        current_balance = Account.loc[account_id, "Balance"]
        new_balance = current_balance + amount
        Account.loc[account_id, "Balance"] = new_balance
        Account.to_csv("Account.csv")
        transaction_id = len(Transaction) + 1
        Transaction.loc[transaction_id] = [account_id, "Deposit", amount]
        Transaction.to_csv("Transaction.csv")
        print(f"Deposit of Rs{amount} completed successfully.")
        print(f"Updated balance for account {account_id}: Rs{new_balance}")
        input("Press Enter to continue: ")

elif transaction_option == 2:
    print("\tWithdrawing Money")
    account_id = int(input("Enter Account ID: "))

    if account_id not in Account.index:
        print("No Such Account Exists")
```

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```python
    if account_id not in Account.index:
        print("No Such Account Exists")
        print("Check the Account ID")
        print("Press Enter to continue: ")

    else:
        amount = float(input("Enter the withdrawal amount (in Rs): "))
        current_balance = Account.loc[account_id, "Balance"]

        if amount > current_balance:
            print("Insufficient amount for withdrawal.")
            input("Press Enter to continue: ")

        else:
            new_balance = current_balance - amount
            Account.loc[account_id, "Balance"] = new_balance
            Account.to_csv("Account.csv")
            transaction_id = len(Transaction) + 1
            Transaction.loc[transaction_id] = [account_id, "Withdrawal", amount]
            Transaction.to_csv("Transaction.csv")
            print(f"Withdrawal of Rs{amount} completed successfully.")
            print(f"Updated balance for account {account_id}: Rs{new_balance}")
            input("Press Enter to continue: ")

elif transaction_option == 3:
    print("\tTransferring Money")
    from_account_id = int(input("Enter Account ID: "))
    to_account_id = int(input("Enter recipient's Account ID: "))

    if from_account_id not in Account.index or to_account_id not in Account.index:
        print("One or both the account IDs do not exist.")
        input("Press Enter to continue: ")

    else:
        amount = float(input("Enter the transfer amount (in Rs): "))
        from_account_balance = Account.loc[from_account_id, "Balance"]

        if amount > from_account_balance:
            print("Insufficient balance for transfer.")
            input("Press Enter to continue: ")
        else:
```

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File Edit Format Run Options Window Help

```python
            input("Press Enter to continue: ")
        else:
            to_account_balance = Account.loc[to_account_id, "Balance"]
            new_from_balance = from_account_balance - amount
            new_to_balance = to_account_balance + amount
            Account.loc[from_account_id, "Balance"] = new_from_balance
            Account.loc[to_account_id, "Balance"] = new_to_balance
            Account.to_csv("Account.csv")
            transaction_id_from = len(Transaction) + 1
            transaction_id_to = len(Transaction) + 2
            Transaction.loc[transaction_id_from] = [from_account_id, "Transfer (to)", amount]
            Transaction.loc[transaction_id_to] = [to_account_id, "Transfer (from)", amount]
            Transaction.to_csv("Transaction.csv")
            print(f"Transfer of Rs{amount} to account {to_account_id} completed successfully.")
            print(f"Updated balance for account {from_account_id}: Rs{new_from_balance}")
            print(f"Updated balance for account {to_account_id}: Rs{new_to_balance}")
            input("Press Enter to continue: ")

elif transaction_option == 4:
    print("\tViewing Transaction History")
    account_id = int(input("Enter Account ID: "))

    if account_id not in Account.index:
        print("No Such Account Exists")
        print("Check the Account ID")
        input("Press Enter to Continue: ")

    else:
        print(f"Transaction History for Account ID: {account_id}")
        print(Transaction[Transaction['Account_ID'] == account_id])
        input("Press Enter to continue: ")

elif choice == 5:
    print("exit")

# For loan csv
elif choice == 4:
    print("          Loan Operations     ")
    print("+----------------------------------------+")
    print("| Option |      Description              |")
    print("+----------------------------------------+")
```

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File Edit Format Run Options Window Help

```python
    print("| Option |     Description              |")
    print("+----------------------------------------+")
    print("|   1    |   Issue a new Loan           |")
    print("|   2    |   View Loan Details          |")
    print("|   3    |   Repay a Loan               |")
    print("+----------------------------------------+")

    loan_option = int(input("Choose your Desired option: "))
    NewCustomer = input("Is this a new customer? \n If yes, press any key and go back to the main menu and first register.")
    if NewCustomer in 'Yy':
        if loan_option == 1:
            print("\tIssuing a New Loan")
            customer_id = input("Enter Customer ID: ")

            # Validate if the customer ID exists
            if customer_id not in Customer.index:
                print("No such customer exists. Please check the customer ID.")
                input("Press Enter to continue: ")

            else:
                try:
                    # Define a predefined loan plan
                    loan_plan = [
                        {"amount": 10000, "term": 6},
                        {"amount": 20000, "term": 12},
                        {"amount": 50000, "term": 18},
                    ]

                    # Set a fixed interest rate
                    monthly_interest_rate = 0.02

                    # Present the predefined loan plan to the user
                    print("\nPredefined Loan Plan: ")
                    print("{:<10} {:<10} {:<10}".format("Amount", "Term(Months)", "Monthly Installment"))
                    for plan in loan_plan:
                        loan_amount = plan["amount"]
                        loan_term = plan["term"]
                        monthly_installment = (
                            loan_amount
                            * (monthly_interest_rate * (1 + monthly_interest_rate) ** loan_term)
                            / ((1 + monthly_interest_rate) ** loan_term - 1)
```

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File Edit Format Run Options Window Help

```python
                            * (monthly_interest_rate * (1 + monthly_interest_rate) ** loan_term)
                            / ((1 + monthly_interest_rate) ** loan_term - 1)
                        )
                        print("{:<10} {:<10} Rs{:<10.2f}".format(loan_amount, loan_term, monthly_installment))

                    # Ask the user if they want to choose a predefined loan plan
                    choice = input("\nDo you want to choose a loan plan from the predefined options? (y/n): ")

                    if choice in "Yy":
                        # Allow the user to choose a plan
                        selected_plan = int(input("Enter the number of loan plan you want to choose: "))

                        if 0 <= selected_plan < len(loan_plan):
                            chosen_plan = loan_plan[selected_plan]
                            loan_amount = chosen_plan["amount"]
                            loan_term = chosen_plan["term"]

                            # Calculate monthly installment
                            monthly_installment = (
                                loan_amount
                                * (monthly_interest_rate * (1 + monthly_interest_rate) ** loan_term)
                                / ((1 + monthly_interest_rate) ** loan_term - 1)
                            )

                            # Save the details to the Dataframe
                            new_loan_details = {
                                "Loan_ID": Loan.index[-1] + 1 if not Loan.empty else 1,
                                "Customer_ID": customer_id,
                                "Loan_Amount": loan_amount,
                                "Interest_Rate": monthly_interest_rate * 12,
                                "Loan_Term": loan_term
                            }

                            # Add a new row to the Dataframe using loc
                            Loan.loc[len(Loan)] = new_loan_details

                            # Save the updated Loan Dataframe to CSV
                            Loan.to_csv("Loan.csv")

                            # Provide information to the user
                            print("\nNew loan issued successfully")
```

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File Edit Format Run Options Window Help

```python
                            # Provide information to the user
                            print("\nNew loan issued successfully")
                            print(f"Loan ID: {new_loan_details['Loan_ID']}")
                            print(f"Loan Amount: Rs{loan_amount}")
                            print(f"Monthly Installment: Rs{monthly_installment:.2f}")
                            print(f"Loan Term: {loan_term} months")
                            input("Press Enter to continue: ")
                        else:
                            print("Invalid Selection. Please choose a valid loan plan")
                            input("Press Enter to continue: ")

                    else:
                        print("\nThank you for considering our loan options.")
                        input("Press Enter to continue: ")

                except ValueError:
                    print("Invalid input")
                    input("Press Enter to continue: ")

        elif loan_option == 2:
            print("\tViewing Loan Details.")
            loan_id = int(input("Enter loan ID: "))

            if loan_id not in Loan.index:
                print("No such loan exists")
                print("Check loan ID")
                input("Press Enter to continue: ")

            else:
                print(Loan.loc[loan_id])
                input("Press Enter to continue: ")

        elif loan_option == 3:
            print("\tRepaying a Loan")
            loan_id = int(input("Enter Loan ID: "))

            if loan_id not in Loan.index:
                print("No such loan exists")
                print("Check loan ID")
                input("Press Enter to continue: ")

```

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```python
            loan_id = int(input("Enter Loan ID: "))

            if loan_id not in Loan.index:
                print("No such loan exists")
                print("Check loan ID")
                input("Press Enter to continue: ")

            else:
                amount_to_repay = float(input("Enter the amount to repay (in Rs): "))
                remaining_balance = Loan.loc[loan_id, "Loan_Amount"] - amount_to_repay

                if remaining_balance < 0:
                    print("Overpayment not Allowed.")
                    print("Press Enter to continue: ")

                else:
                    Loan.loc[loan_id, "Loan_Amount"] = remaining_balance
                    Loan.to_csv("Loan.csv")
                    print(f"Repayment of Rs{amount_to_repay} completed successfully.")
                    print(f"Remaining balance for loan {loan_id}: Rs{remaining_balance}")
                    input("Press Enter to continue: ")

        else:
            print("Go to the Main Menu and Register details.")
elif choice == 5:
    print("\tCustomer Care details")
    print("For any assistance, please contact our customer care: ")
    print("Phone: 1800-ABC-BANK (1800-342343-2265)")
    print("Email: customercare@abcbank.com")
    print("Address: ABC Bank, 123 Main Street, Jaipur, India")
    input("Press Enter to continue: ")

elif choice == 6:
    print("Thank You!")

else:
    print("Invalid user ID or password. Please check again.")
```

OUTPUT

Introduction screen

```
*IDLE Shell 3.13.1*
File  Edit  Shell  Debug  Options  Window  Help
Enter your user ID: 1
Enter your Password: 123
Hello Bank Manager
Press any key to continue
+--------------------------------------------------+
|                    Welcome                       |
|               ********************               |
|                   MENU LIST                      |
+__________________________________________________+
+____________________________________+
| 0     | Introduction               |
| 1     | Account operation          |
| 2     | Customer operation         |
| 3     | Transaction operation      |
| 4     | Loan operation             |
| 5     | Customer care details      |
| 6     | Exit                       |
+____________________________________+
Enter any digit number from the above list: 0
+____________________________________________________________________________________________________+
|Introduction to bank management                                                                     |
+____________________________________________________________________________________________________+

                                Bank management involves overseeing a range of activities to ensure the smooth functioning and success of a bank. This includes st
rategic planning, financial management, risk assessment, regulatory compliance, customer service, human resource management, technology integration, marketing, cor
porate governance, and financial reporting.
The management team is responsible for setting long-term goals, making decisions on financial resources, mitigating risks, complying with regulations, enhancing cu
stomer satisfaction, managing human resources, adopting technology, promoting the bank's image, and maintaining transparency.
Effective bank management is vital for the stability and growth of a financial institution.
 The main objective is to build an organic and optimal interaction system.
+____________________________________________________________________________________________________+
+____________________________________+
| 0     | Introduction               |
| 1     | Account operation          |
| 2     | Customer operation         |
| 3     | Transaction operation      |
| 4     | Loan operation             |
| 5     | Customer care details      |
| 6     | Exit                       |
+____________________________________+
Enter any digit number from the above list:
```

Menu list display

```
Enter your user ID: 1
Enter your Password: 123
Hello Bank Manager
Press any key to continue
+--------------------------------------------------+
|                    Welcome                       |
|               ********************               |
|                   MENU LIST                      |
+__________________________________________________+
+____________________________________+
| 0     | Introduction               |
| 1     | Account operation          |
| 2     | Customer operation         |
| 3     | Transaction operation      |
| 4     | Loan operation             |
| 5     | Customer care details      |
| 6     | Exit                       |
+____________________________________+
Enter any digit number from the above list: 2
Customer Operations
+-------------------------------+
|OPTION|      DESCRIPTION       |
+-------------------------------+
|  1   | Add a new customer      |
|  2   | Modify an existing customer |
|  3   | Remove a customer       |
|  4   | View customer information   |
|  5   | Graphical representation |
|  6   | Exit                    |
+-------------------------------+
Choose your desired option:
```

Modifying an existing customer

```
Want to print a Specific customer press 1
Want to print Top 5 customer details press 2
Want to print bottom 5 customer details press 3
Want to print only customer names and Phone numbers press 4
Want to print customer names in ascending order press 5
Want to print customer names in descending order press 6
Enter your choice: 1
>>>
```

Transaction operations

```
Enter any digit from the above list: 3
        Transaction Operations
+----------------------------------------+
| Option |     Description               |
+----------------------------------------+
|   1    |   Deposit                     |
|   2    |   Withdraw                    |
|   3    |   Transfer                    |
|   4    |   View Transaction History    |
|   5    |   Exit                        |
+----------------------------------------+
Choose your Desired Option: 3
Transferring money
Enter account ID: 1013
```

Viewing customer information

```
Want to print a Specific customer press 1
Want to print Top 5 customer details press 2
Want to print bottom 5 customer details press 3
Want to print only customer names and Phone numbers press 4
Want to print customer names in ascending order press 5
Want to print customer names in descending order press 6
Enter your choice: 2
0      john
1    olivia
2     gracy
3     simon
dtype: object
Press enter to continue.
```

```
Want to print a Specific customer press 1
Want to print Top 5 customer details press 2
Want to print bottom 5 customer details press 3
Want to print only customer names and Phone numbers press 4
Want to print customer names in ascending order press 5
Want to print customer names in descending order press 6
Enter your choice: 3
2    gracy
3    simon
4      Ben
5      ram
6    sohan
dtype: object
Press enter to continue.
```

Customer operations:

```
Choose your Desired Option: 4
Viewing transaction history
Enter account ID: 1013
Transaction History for Account ID: 1013
Transaction ID          Account ID          Transaction Type        Amount
25                      1013                Deposit                 1000
26                      1013                Withdrawl               1000
28                      1013                Transfer(from)          1000
Press enter to continue:
```

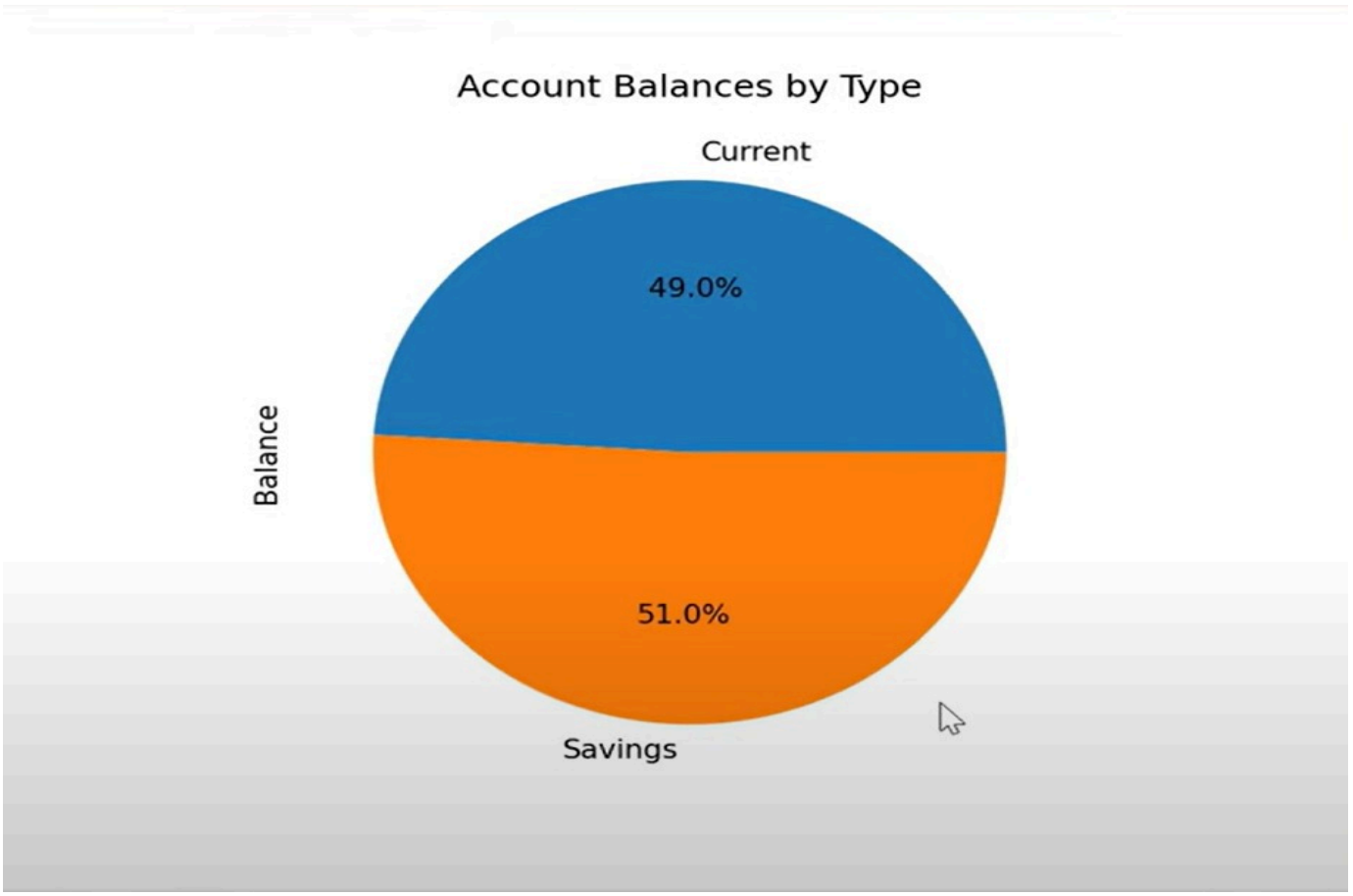
Loan operations

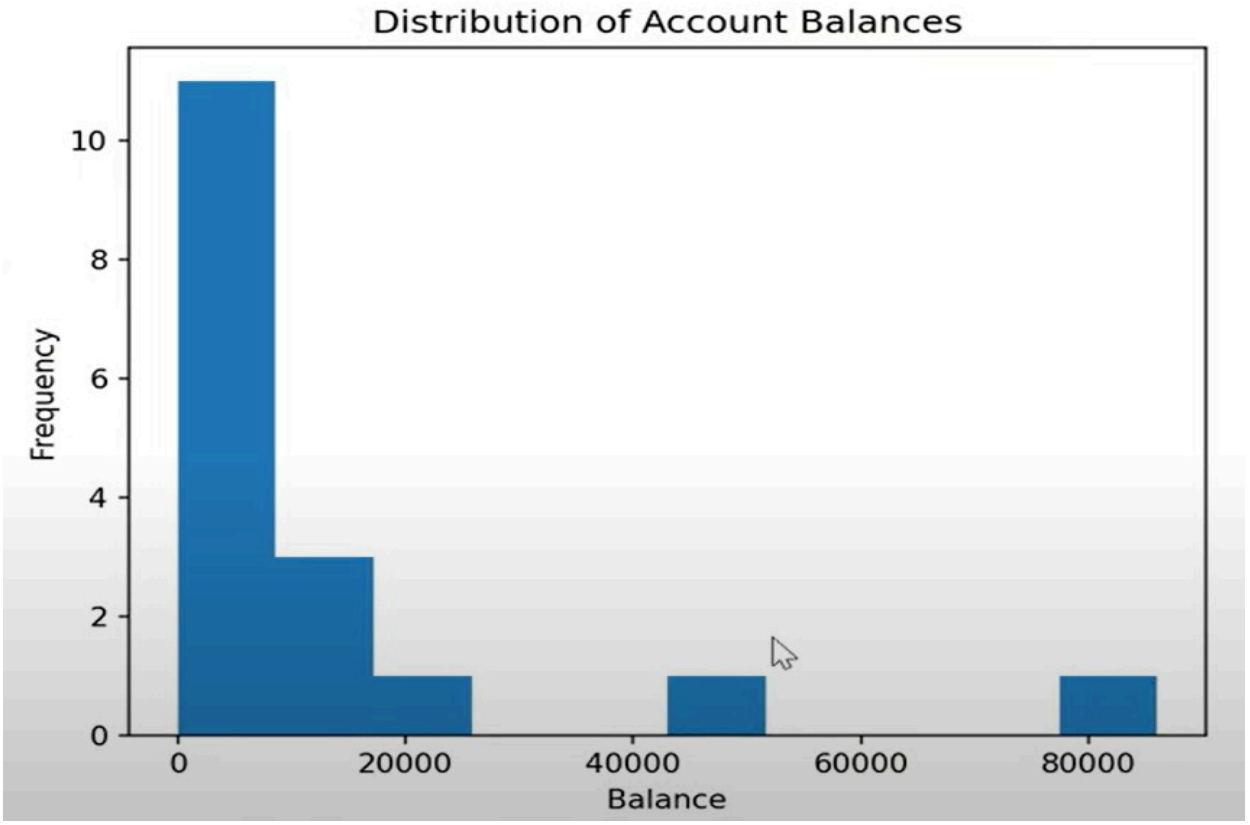
```
Enter a digit from the above list:
         Loan Operations
+----------------------------------------+
| Option |     Description               |
+----------------------------------------+
|   1    |   Issue a new Loan            |
|   2    |   View Loan Details           |
|   3    |   Repay a Loan                |
+----------------------------------------+
Choose your Desired option: 1
Is this a new customer?
 If yes, press any key and go back to the main menu and first register.
Issuing a new loan
Enter Customer ID: C1014
Predefined loan plan:
Amount          Term(months)          Monthly Installment
10000               6                     1786.6
20000               8                     1985.6
500000              12                    2005.7
Press enter to continue:
```

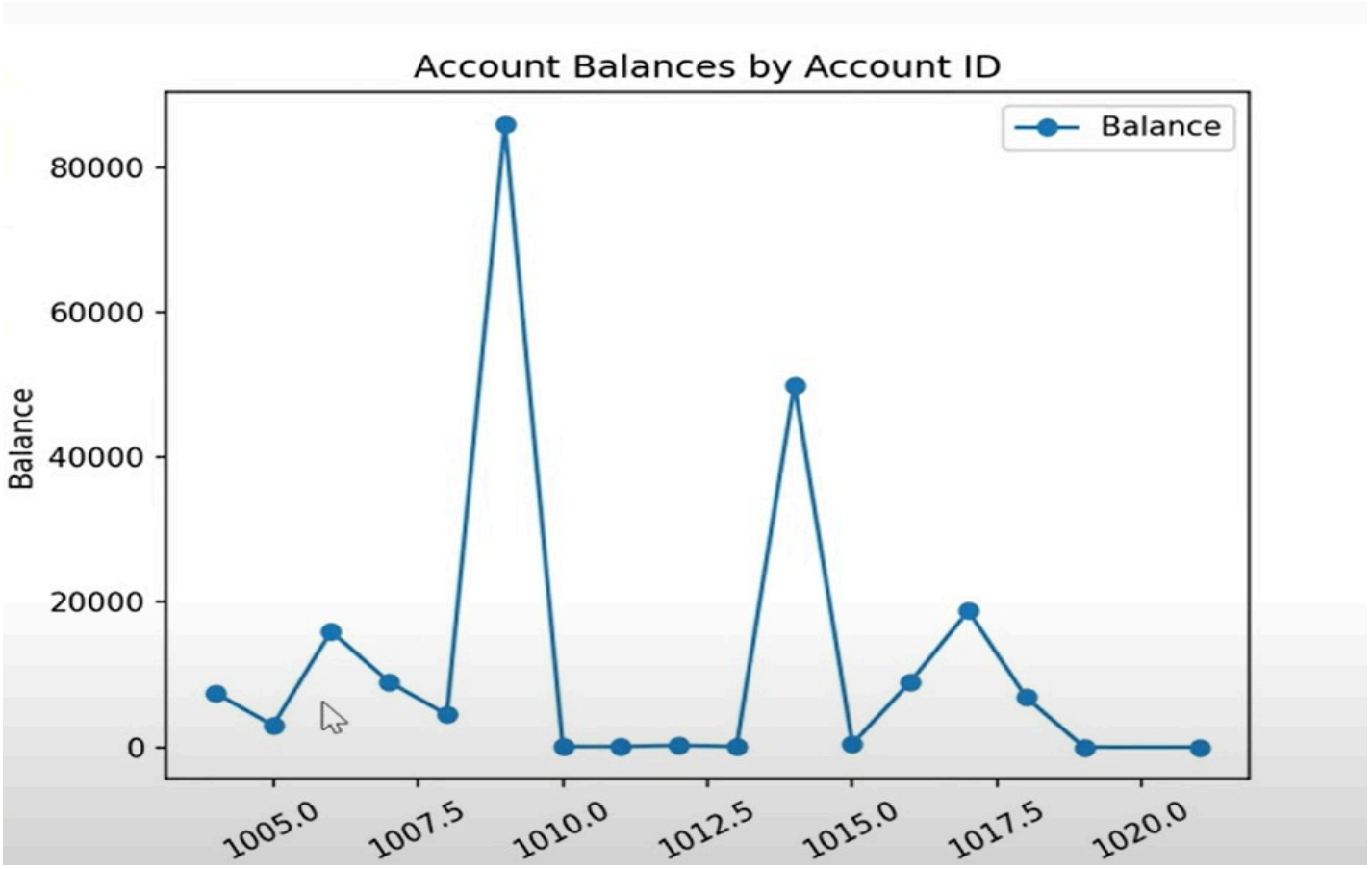
GRAPHICAL REPRESENTATION:

```
Graphical representation of account data
+---------------------------------------+
|OPTION|      DESCRIPTION               |
+---------------------------------------+
|  1   | Bar Chart: Total balance by type
|  2   | Pie Chart: Balance distribution
|  3   | Line Chart: Balance over time
|  4   | Histogram: Balance distribution
+---------------------------------------+
>>>
```

PIE CHART DISTRIBUTION




Distribution of Account Balances
Frequency
10
8
6
4
2
0
0
20000
40000
60000
80000
Balance


Account Balances by Account ID
Balance
80000
60000
40000
20000
0
Balance
1005.0
1007.5
1010.0
1012.5
1015.0
1017.5
1020.0

CSV FILES

Account.csv

Account_I	Customer	Account_	Balance
1004	Ronak	Current	7500
1005	Rishi	Savings	3000
1006	Piyush	Current	16000
1007	Saksham	Savings	9000
1008	Ravi	Current	4500
1009	Priya	Savings	86000
1010	Shubham	Savings	100
1011	Garv	Savings	100
1012	Purvansh	Savings	250
1013	Aman	Savings	100
1014	Anjali	Current	50000
1015	Ram	Savings	450
1016	Ram	Savings	9000
1017	Iuuj	Current	18801
1018	Ram	Current	6865
1019	Ram	Savings	90
1020	Op Tiger	Savings	987

loan.csv

Customer_ID, Customer_Name, Customer_Address, Customer_Phone, Customer_Email C1002, Jane Smith, 456 Elm St, (555) 555-5678, jane.smith@email.com C1003, Komal, ert, 34, er
C1004, Alice Brown, 101 Pine St, (555) 555-3456, alice.brown@email.com C1005, Charlie Lee, 202 Maple St, (555) 555-7890, charlie.lee@email.com
C1006, Eve Taylor, 303 Cedar St, (555) 555-2345, eve.taylor@email.com
C1008, Grace Anderson, 505 Redwood St, (555) 555-1234, grace. anderson@email.com C1009, Francis Garcia, 606 Fir St, (555) 555-5678, francis.garcia@email.com
C1010, Heather Martinez, 707 Sycamore St, (555) 555-9012, heather.martinez@email.com C1011, ram, sitapura, 2598631475, ram@gmail.com
C1012, Op Singh, 12 Lane Fatehput Sikri, 343453453, op@gmail.com
C1013, Op tiger, rota, 3224324, sdfsd
c1014, Topi, Kota, 2324324324, op@gmail.com

Transaction.csv

Transaction_ID, Account_ID, Transaction_Type, Amount 1,1001, Deposit, 1000.0
2,1002, Withdrawal,500.0 3,1003, Deposit, 1500.0
4,1004, Withdrawal, 1000.0
5,1005, Deposit,800.0
6,1006, withdrawal,300.0 7,1007, Deposit, 2000.0
8,1008, Withdrawal, 1200.0 9,1009, Deposit,950.0 10,1010, Withdrawal,600.0 11,1003, Deposit, 30000.0 12,1003, Deposit, 90000.0 13,1003, Withdrawal, 10000.0 14,1003, Transfer (to), 1000.0 15,1007, Transfer (from), 100
18,1009, Transfer (to), 10000.0 19,1006, Transfer (from), 10000.0
20,1017, Deposit, 12000.0
21,1018, Withdrawal,123.0
22,1018, Transfer (to), 12.0
23,1017, Transfer (from), 12.0
24,1009, Deposit, 1200.0 25,1021, Deposit, 1000.0
26,1021, Withdrawal, 1000.0 27,1011, Transfer (to), 12.0

Customer.csv

Transaction_ID, Account_ID, Transaction_Type, Amount 1,1001, Deposit, 1000.0
2,1002, Withdrawal,500.0 3,1003, Deposit, 1500.0
4,1004, Withdrawal, 1000.0
5,1005, Deposit,800.0
6,1006, withdrawal,300.0 7,1007, Deposit, 2000.0
8,1008, Withdrawal, 1200.0 9,1009, Deposit,950.0 10,1010, Withdrawal,600.0 11,1003, Deposit, 30000.0 12,1003, Deposit, 90000.0 13,1003, Withdrawal, 10000.0 14,1003, Transfer (to), 1000.0 15,1007, Transfer (from), 100
18,1009, Transfer (to), 10000.0 19,1006, Transfer (from), 10000.0
20,1017, Deposit, 12000.0
21,1018, Withdrawal,123.0
22,1018, Transfer (to), 12.0
23,1017, Transfer (from), 12.0
24,1009, Deposit, 1200.0 25,1021, Deposit, 1000.0
26,1021, Withdrawal, 1000.0 27,1011, Transfer (to), 12.0

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